Midterm Examination: Solutions and Grading Guide

Prof. J. Whitfield

Notes for graders. Reward correct strategic logic. A student who makes an early arithmetic mistake and then reasons correctly from it loses the points for the mistake only. Strategies must be *complete* plans (an action at every decision node, including nodes off the equilibrium path); stating only the outcome of an equilibrium earns at most half of the points for that part.

Problem 1 [32 points]

(a) Pacific Air moves at three decision nodes (the root, after (N, In), after (E, In)) with two actions each: $2^3 = 8$ pure strategies. SkyLink moves at two nodes (after N and after E): $2^2 = 4$ pure strategies. Writing (action after N, action after E), they are

$$(In, In), \quad (In, Out), \quad (Out, In), \quad (Out, Out).$$

Grading (4 points).

- 1 pt each for the counts 8 and 4. 2 pts: SkyLink's strategies listed as contingent plans.
- Answering "Pacific has 2 strategies" (only the first move): 0 for that count.

(b) Strategic form of the subgame after N (Pacific rows, SkyLink columns):

	In	*Out*
F	2, -3	12, 0
A	6, 2	12, 0

Pure Nash equilibria: (A, In) with payoffs $(6, 2)$, and (F, Out) with payoffs $(12, 0)$. In (F, Out) Pacific is indifferent between F and A (neither is reached), and SkyLink prefers *Out* ($0 > -3$) given the threat. The equilibrium (F, Out) relies on a non-credible threat: if SkyLink actually entered, Pacific would earn 2 by fighting but 6 by accommodating, so it would not carry out the threat.

Grading (6 points).

- 2 pts: correct matrix. 2 pts: both equilibria (1 each).
- 2 pts: identify (F, Out) as the non-credible one with the comparison $2 < 6$.

(c) Backward induction:

- After (N, In): A ($6 > 2$). After (E, In): F ($5 > 2$).
- SkyLink after N: *In* yields 2 (Pacific accommodates) > 0, so *In*. After E: *In* yields -3 (Pacific fights) < 0, so *Out*.
- Pacific at the root: N leads to $(6, 2)$, E leads to $(9, 0)$; choose E.

SPE: Pacific plays (E; A after (N, In); F after (E, In)). SkyLink plays (In after N; Out after E). Outcome: Pacific expands, SkyLink stays out; payoffs $(9, 0)$.

Grading (8 points).

- 2 pts: Pacific's two last-stage choices. 2 pts: SkyLink's two choices.
- 2 pts: Pacific's root choice with comparison $9 > 6$.
- 2 pts: complete strategies (including off-path actions) and the outcome.

(d)(i) After E, fighting yields $8-k$ and accommodating $5-k$, so Pacific fights for every k; SkyLink therefore stays out after E, and Pacific gets $12-k$. Without expansion it gets 6. Pacific expands iff $12-k \geq 6$, i.e., $k \leq 6$.

(d)(ii) With $k = 3$, Pacific fights after (E, In) iff $F_g - 3 > 2$, i.e., $F_g > 5$. Then SkyLink stays out and Pacific earns $9 > 6$, so it expands and deters entry. If $F_g < 5$, Pacific would accommodate after (E, In), SkyLink would enter after E, and expanding would yield only $2 < 6$; so Pacific does not expand, SkyLink enters and Pacific accommodates (payoffs $(6, 2)$). (At $F_g = 5$ Pacific is indifferent after (E, In), and whether expansion deters depends on how the tie is resolved.) The lesson: expansion deters only if it changes Pacific's *own* post-entry incentive to fight; its cost only determines whether deterrence is worth buying.

Grading (6 points).

- (i) 3 pts: fighting after E is independent of k (1), comparison $12-k$ vs. 6 (1), $k \leq 6$ (1).
- (ii) 3 pts: threshold $F_g > 5$ (2), the SPE when $F_g < 5$ (1).

(e) Consider Pacific: (N; F after (N, In); F after (E, In)) and SkyLink: (Out after N; Out after E). The outcome is (N, Out) with payoffs $(12, 0)$.

- Pacific: switching to E yields $9 < 12$ (SkyLink still stays out); changing its responses at unreached nodes does not change its payoff.
- SkyLink: after N, entering would yield -3 (Pacific fights) < 0; its action after E is not reached.

So this is a Nash equilibrium. It is not subgame perfect: in the subgame after (N, In) Pacific would earn 2 from F and 6 from A, so the threat to fight is not credible, and SkyLink should not believe it.

Pacific pays for capacity in the SPE because expansion is a *commitment device*. Without it, a rational entrant knows the fare war would not happen and enters, leaving Pacific with 6. The expansion changes Pacific's own payoffs (fighting then yields $5 > 2$), which makes the threat credible, so SkyLink stays out. The payoff of 12 is available only if SkyLink believes an incredible threat; among outcomes consistent with sequential rationality, the best Pacific can do is pay 3 for credibility and earn 9 instead of 6.

Grading (8 points).

- 2 pts: complete strategy profile. 2 pts: verify both firms' best responses.
- 2 pts: explain why not SPE (non-credible F after (N, In)).
- 2 pts: commitment/credibility explanation of the expansion.

Problem 2 [28 points]

(a) L strictly dominates H for both airlines: Pacific $14 > 10$ and $5 > 2$; SkyLink $9 > 6$ and $3 > 1$. The unique Nash equilibrium is (L, L) with payoffs $(5, 3)$. It is a prisoner's dilemma because both airlines have

a dominant strategy, yet the resulting outcome is Pareto-dominated by (H,H) with payoffs $(10,6)$.

Grading (4 points).

- 2 pts: (L,L) via dominance. 2 pts: PD explanation (dominant strategies and Pareto inferiority).

(b) In quarter 8, whatever happened before, the airlines play the stage game once, so they play its unique Nash equilibrium (L,L). In quarter 7, play in quarter 8 does not depend on what happens in quarter 7, so only current profits matter, and again both play L. By induction, the unique SPE has both airlines setting L in every quarter after every history. Payoffs: 5 and 3 per quarter.

Grading (4 points).

- 2 pts: last-period argument. 2 pts: induction and uniqueness (the unique stage NE is essential).

(c) After a deviation, both play (L,L) forever, which repeats the stage Nash equilibrium and is therefore a Nash equilibrium of every such subgame. In the cooperative phase, by the one-shot deviation principle, it suffices to compare cooperating forever with deviating once and being punished forever.

$$\text{Pacific: } \frac{10}{1-\delta} \;\geq\; 14+\frac{5\delta}{1-\delta} \iff 10 \geq 14-9\delta \iff \delta \geq \tfrac{4}{9}.$$

$$\text{SkyLink: } \frac{6}{1-\delta} \;\geq\; 9+\frac{3\delta}{1-\delta} \iff 6 \geq 9-6\delta \iff \delta \geq \tfrac{1}{2}.$$

Grim trigger is an SPE iff $\delta \geq \frac{1}{2}$; SkyLink's constraint binds. In general the threshold is $\delta^* = \frac{\pi^{\text{dev}}-\pi^{\text{coop}}}{\pi^{\text{dev}}-\pi^{\text{pun}}}$: Pacific's is $\frac{4}{9}$ and SkyLink's is $\frac{3}{6}$. For SkyLink, cooperation is worth only 3 per quarter more than the price war (6 vs. 3), while undercutting yields an immediate gain equal to that full amount (9 vs. 6). The smaller carrier has relatively the most to gain from undercutting and the least to lose from a fare war, so it is the harder one to keep in line.

Grading (8 points).

- 1 pt: punishment phase is a stage-NE repetition (credible).
- 2 pts each: the two incentive constraints with correct thresholds.
- 3 pts: overall threshold $\delta \geq 1/2$ with SkyLink's constraint binding (1 pt) and a sensible economic explanation (2 pts).
- Students who find only one airline's threshold: at most 4 of 8.

(d)(i) Future profits are received only if the route survives, so the effective discount factor is $0.8\,\delta$. The binding (SkyLink) condition becomes $0.8\,\delta \geq \frac{1}{2}$, i.e., $\delta \geq \frac{5}{8} = 0.625$ (Pacific's becomes $\delta \geq \frac{5}{9}$, which is weaker).

(d)(ii) $e^{-r/4} \geq \frac{5}{8} \iff r \leq 4\ln\frac{8}{5} \approx 1.88$, i.e., a continuously compounded rate of about 188% per year. At $r = 10\%$, $\delta = e^{-0.025} \approx 0.975 > 0.625$, so cooperation is easily sustainable; what matters in practice is the closure risk and how often the airlines interact, not the interest rate.

Grading (6 points).

- (i) 3 pts: effective discount factor 0.8δ (2), threshold $5/8$ (1).
- (ii) 3 pts: bound $r \leq 4\ln(8/5)$ (2), conclusion at 10% (1).

(e) $m = 1$. SkyLink's payoffs from deviating and then returning to cooperation are $9, 3, 6, 6, \ldots$ versus $6, 6, 6, \ldots$ from cooperating. Only the first two quarters differ, so it deviates iff $9 + 3\delta > 6 + 6\delta$, i.e., $3 > 3\delta$, which holds for every $\delta < 1$.

$m = 2$. Only the first three quarters differ.

$$\text{SkyLink: } 9 + 3\delta + 3\delta^2 \leq 6 + 6\delta + 6\delta^2 \iff \delta^2 + \delta - 1 \geq 0 \iff \delta \geq \frac{\sqrt{5} - 1}{2} \approx 0.618.$$

$$\text{Pacific: } 14 + 5\delta + 5\delta^2 \leq 10 + 10\delta + 10\delta^2 \iff \delta^2 + \delta - 0.8 \geq 0 \iff \delta \geq \frac{\sqrt{105} - 5}{10} \approx 0.525.$$

Neither airline deviates iff $\delta \geq \frac{\sqrt{5}-1}{2} \approx 0.618$. A milder punishment requires more patience than grim trigger ($0.618 > 0.5$).

Grading (6 points).

- 2 pts: $m = 1$ argument for SkyLink.
- 2 pts: SkyLink's $m = 2$ condition and threshold 0.618.
- 1 pt: Pacific's $m = 2$ threshold 0.525. 1 pt: overall threshold and comparison with grim trigger.